feb. 17 Ch.4: Classifying Cyclic Groups

$\mathbb{C}$ complex plane

Cycle around the complex unit circle, hence the name cyclic

$x^n - 1 = 0$ $\quad \langle i \rangle = \{i^n : n \in \mathbb{Z}\} = \{i, -1, -i, 1\}$

$x^n = 1$ $\quad$ ★ won't be tested on this, just context for name cyclic

- Classifying cyclic groups is part of larger program of classifying groups
- Class spent on worksheet (on Canvas), condensed results of Ch.4 with some proofs and blanks to be filled in
  - Worked in groups/collaboratively, only subgroup diagram at the end done as a class
  - Not all of worksheet done today, check textbook for further reading & filling in blanks

Chapter 4 Results as a Worksheet

Theorem 4.1 Criterion for $a^i = a^j$

Let G be a group, and let a belong to G. If a has infinite order, then $a^i = a^j$ if and only if $i = j$. If a has finite order, say, n, then $\langle a \rangle = \{e, a, a^2, \ldots, a^{n-1}\}$ and $a^i = a^j$ if and only if n divides $i - j$.

PROOF If a has infinite order, there is no nonzero n such that a^n is the identity. Since $a^i = a^j$ implies $a^{i-j} = e$, we must have $i - j = 0$, and the first statement of the theorem is proved. $a^i = a^j \Rightarrow a^i a^{-j} = a^j a^{-j} \Rightarrow a^i a^{-j} = e$

Now assume that $|a| = n$. We will prove that $\langle a \rangle = \{e, a, \ldots, a^{n-1}\}$. Certainly, the elements $e, a, \ldots, a^{n-1}$ are in $\langle a \rangle$. $\Rightarrow a^{i-j} = e$, $a^{i-j} = a^0$ so $i-j=0$

Now, suppose that a^k is an arbitrary member of $\langle a \rangle$. By the division algorithm, there exist integers q and r such that

$$k = qn + r \quad \text{with} \quad 0 \le r < n.$$

Since $a^k = a^r$ and $0 \le r < n$, and a^0 through a^{n-1} are in it

Then $a^k = a^{qn+r} = a^{qn}a^r = (a^n)^q a^r = ea^r = a^r$, so that $a^k \in \{e, a, a^2, \ldots, a^{n-1}\}$. This proves that $\langle a \rangle = \{e, a, a^2, \ldots, a^{n-1}\}$.

Next, we assume that $a^i = a^j$ and prove that n divides $i - j$. We begin by observing that $a^i = a^j$ implies $a^{i-j} = e$. Again, by the division algorithm, there are integers q and r such that

$$i - j = qn + r \qquad \text{with} \qquad 0 \le r < n.$$

What should the argument be in the redacted spot?

$a^{i-j} = a^{qn+r} = a^{qn}a^r = (a^n)^q a^r = (a^n)^q a^r = (e)^q a^r$

$ea^r = a^r$. Since n is the least positive integer such that a^n is the identity, we must have $r = 0$, so that n divides $i - j$. Since we said $a^{i-j} = e$, and $a^{i-j} = a^r$ so $i-j = 0 = r$

Conversely, if $i - j = nq$, then $a^{i-j} = a^{nq} = e^q = e$, so that $a^i = a^j$. ∎

Corollary 1 $|a| = |\langle a \rangle|$

For any group element a, $|a| = |\langle a \rangle|$.

Corollary 2 $a^k = e$ Implies That $|a|$ Divides k

Let G be a group and let a be an element of order n in G. If $a^k = e$, then n divides k.

Corollary 3 Relationship between $|ab|$ and $|a||b|$

Write a proof for Corollary 3.

If a and b belong to a finite group and $ab = ba$, then $|ab|$ divides $|a||b|$.

Let $|a|=n$ and $|b|=m$. Observe that $(ab)^{nm} = a^{nm}b^{nm} = (a^n)^m(b^m)^n = (a^n)^m(b^m)^n = (e)^m(e)^n = e$. So $(ab)^{nm} = e$. Since $|a||b| = nm$, if $|ab| = nm$, then $|ab| = |a||b|$ and $|ab|$ divides $|a||b|$. However, nm may not necessarily be the smallest positive integer that satisfies the order condition, but by Cor. 2, it is a multiple of it. Call $|ab| = \ell$. Then, $nm = \ell \cdot p$ for some integer p. Thus, $\ell \mid nm \Rightarrow |ab|$ divides $|a||b|$. □

Theorem 4.2 $\langle a^k \rangle = \langle a^{\gcd(n,k)} \rangle$ and $|a^k| = n/\gcd(n, k)$

Let a be an element of order n in a group and let k be a positive integer. Then $\langle a^k \rangle = \langle a^{\gcd(n,k)} \rangle$ and $|a^k| = n/\gcd(n, k)$.

What should be in the redacted boxes?

double containment to show groups are equal

PROOF To simplify the notation, let $d = \gcd(n, k)$ and let $k = dr$. Since $a^k = (a^d)^r$, we have by closure that $\langle a^k \rangle \subseteq \langle a^d \rangle$. By Theorem 0.2 (the gcd theorem), there are integers s and t such that $d = ns + kt$. So, $a^d = a^{ns+kt} = a^{ns}a^{kt} = (a^n)^s(a^k)^t = e(a^k)^t = (a^k)^t \in \langle a^k \rangle$. This proves $\langle a^d \rangle \subseteq \langle a^k \rangle$. So, we have verified that $\langle a^k \rangle = \langle a^{\gcd(n,k)} \rangle$.

We prove the second part of the theorem by showing first that $|a^d| = n/d$ for any divisor d of n. Clearly, $(a^d)^{n/d} = a^n = e$, so that $|a^d| \leq n/d$. On the other hand, if i is a positive integer less than n/d, then $(a^d)^i \neq e$ by definition of $|a|$. We now apply this fact with $d = \gcd(n, k)$ to obtain $|a^k| = |\langle a^k \rangle| = |\langle a^{\gcd(n,k)} \rangle| = |a^{\gcd(n,k)}| = n/\gcd(n, k)$. ∎

Corollary 1 Orders of Elements in Finite Cyclic Groups

Prove Corollary 1.

In a finite cyclic group, the order of an element divides the order of the group.

Corollary 2 Criterion for $\langle a^i \rangle = \langle a^j \rangle$ and $|a^i| = |a^j|$

Let $|a| = n$. Then $\langle a^i \rangle = \langle a^j \rangle$ if and only if $\gcd(n, i) = \gcd(n, j)$, and $|a^i| = |a^j|$ if and only if $\gcd(n, i) = \gcd(n, j)$.

Corollary 3 Generators of Finite Cyclic Groups

Let $|a| = n$. Then $\langle a \rangle = \langle a^j \rangle$ if and only if $\gcd(n, j) = 1$, and $|a| = |\langle a^j \rangle|$ if and only if $\gcd(n, j) = 1$.

Corollary 4 Generators of Z_n

An integer k in Z_n is a generator of Z_n if and only if $\gcd(n, k) = 1$.

Explain why Corollary 4 follows from the previous corollaries. Find all generators of Z/15Z.

Theorem 4.3 Fundamental Theorem of Cyclic Groups

What should be in the redacted boxes?

Every subgroup of a cyclic group is cyclic. Moreover, if $|\langle a\rangle| = n$, then the order of any subgroup of $\langle a\rangle$ is a divisor of n; and, for each positive divisor k of n, the group $\langle a\rangle$ has exactly one subgroup of order k—namely, $\langle a^{n/k}\rangle$.

PROOF Let $G = \langle a\rangle$ and suppose that H is a subgroup of G. We must show that H is cyclic. If it consists of the identity alone, then clearly H is cyclic. So we may assume that $H \neq \{e\}$. We now claim that H contains an element of the form a^t, where t is positive. Since $G = \langle a\rangle$, every element of H has the form [redacted] Now let m be the least positive integer such that $a^m \in H$. By closure, $\langle a^m\rangle \subseteq H$. We next claim that $H = \langle a^m\rangle$. To prove this claim, it suffices to let b be an arbitrary member of H and show that b is in $\langle a^m\rangle$. Since $b \in G = \langle a\rangle$, we have $b = a^k$ for some k. Now, apply the division algorithm to k and m to obtain integers q and r such that $k = mq + r$ where $0 \leq r < m$. Then $a^k = a^{mq+r} = a^{mq}a^r$, so that $a^r = a^{-mq}a^k$. Since $a^k = b \in H$ and $a^{-mq} = (a^m)^{-q}$ is in H also, $a^r \in H$. But, m is [redacted] must be 0. Therefore, $b = a^k = a^{mq} = (a^m)^q \in \langle a^m\rangle$. This proves the assertion of the theorem that every subgroup of a cyclic group is cyclic.

To prove the next portion of the theorem, suppose that $|\langle a\rangle| = n$ and H is any subgroup of $\langle a\rangle$. We have already shown that $H = \langle a^m\rangle$, where m is the least positive integer such that $a^m \in H$. Using $e = b = a^n$ as in the preceding paragraph, we have $n = mq$.

Finally, let k be any positive divisor of n. We will show that $\langle a^{n/k}\rangle$ is the one and only subgroup of $\langle a\rangle$ of order k. From Theorem 4.2, we see that $\langle a^{n/k}\rangle$ has order $n/\gcd(n, n/k) = n/(n/k) = k$. Now let H be any subgroup of $\langle a\rangle$ of order k. We have already shown above that $H = \langle a^m\rangle$, where m is a divisor of n. Then $m = \gcd(n, m)$ and $k = |a^m| = |a^{\gcd(n,m)}| = n/\gcd(n, m) = n/m$. Thus, $m = n/k$ and $H = \langle a^{n/k}\rangle$. ∎

Corollary Subgroups of Z_n

For each positive divisor k of n, the set $\langle n/k\rangle$ is the unique subgroup of Z_n of order k; moreover, these are the only subgroups of Z_n.

Theorem 4.4 Number of Elements of Each Order in a Cyclic Group

If d is a positive divisor of n, the number of elements of order d in a cyclic group of order n is $\phi(d)$.

The function ϕ is called Euler's phi function or Euler's totent function. $\phi(n)$ is defined to be the number of integers less than n that are relatively prime to n (i.e., have gcd 1 with n). For example, $\phi(6)=2$. Verify Theorem 4.4 for the cyclic group Z/9Z.

Corollary Number of Elements of Order d in a Finite Group

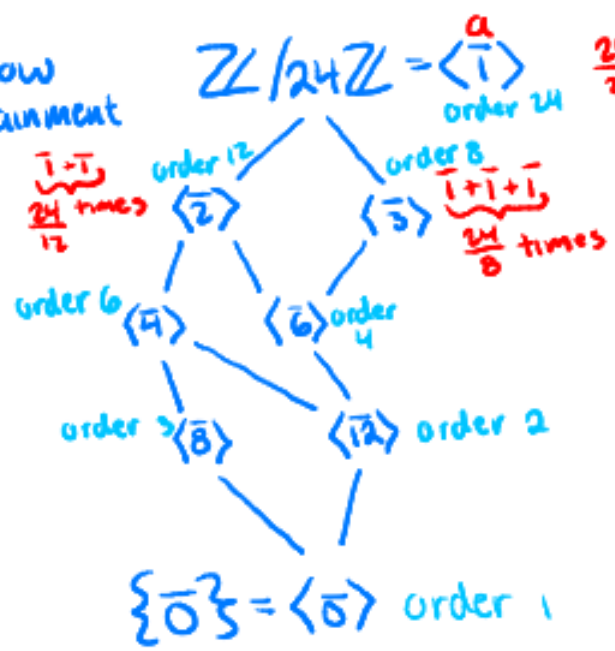
In a finite group, the number of elements of order d is a multiple of $\phi(d)$.

PROOF If a finite group has no elements of order d, the statement is true, since $\phi(d)$ divides 0. Now suppose that $a \in G$ and $|a| = d$. By Theorem 4.4, we know that $\langle a \rangle$ has $\phi(d)$ elements of order d. If all elements of order d in G are in $\langle a \rangle$, we are done. So, suppose that there is an element b in G of order d that is not in $\langle a \rangle$. Then, $\langle b \rangle$ also has $\phi(d)$ elements of order d. This means that we have found $2\phi(d)$ elements of order d in G provided that $\langle a \rangle$ and $\langle b \rangle$ have no elements of order d in common. If there is an element c of order d that belongs to both $\langle a \rangle$ and $\langle b \rangle$, then we have $\langle a \rangle = \langle c \rangle = \langle b \rangle$, so that $b \in \langle a \rangle$, which is a contradiction. Continuing in this fashion, we see that the number of elements of order d in a finite group is a multiple of $\phi(d)$. ∎

Does this proof look familiar? Where have you seen it before?

Use Theorem 4.3 to draw a complete subgroup diagram for Z/24Z. What would you have to change if you were simply asked for a complete subgroup diagram of an abstract cyclic group of order 24?

lines show proper containment



$\frac{24}{24} \cdot \bar{1} = \bar{1}$ group operation w/ itself 24/24 times

divisors of 24
1, 2, 3, 4, 6, 8, 12, 24

"$a^{n/k}$" $= \frac{n}{k} \cdot a$

Since this is an additive group